

COSC480 – Computer Programming

A1. Project part 1 – Read me

Attribute	Details
Individual/Group	Individual
Length	500 words (+/- 10%)
Course Learning Outcomes (Relevant to this assessment)	The course Learning Outcomes demonstrated by successful completion of the task below include: • Develop a program to cleanse data, including organising, identify trends and visualise data for ill-specified, complex data-science problems. Note: This assignment will start the journey to achieving this learning outcome, but it will not be fully achieved until the end of A3. Project part 3 - Final project.
Conditions	• This is a compulsory assignment and non-submission will result in failing the course. No re-sits or resubmissions are allowed. • Any sources used should be cited and referenced in APA 7th format. References will not be counted in the word count for the assignment. • You must not host your project on any public repository. (Please review Git Exercises in Module 1: Introducing Python if this is not clear.)
Submission	Due by 11:55 pm Sunday of Module 2.
Weighting	5%
Maximum grade	13

Assessment task

In the final project for this course, you will be expected to write a program to cleanse, analyse, visualise and, identify trends for the provided dataset as well as additional data that you will source.

This first assessment provides a foundation for the final project by providing you with a dataset and starting point for a program to build on, which will continue in Assessment 2.

First, you'll be introduced to a version control system called Git. Git allows you to track and manage changes to software code in chunks called commits. In this case, Git will be used to track changes to the code for the program you will develop throughout this course. Initial instructions for accessing, installing, and using Git can be found in the Software installation section of the course.

This assignment will require you to make your first commit, while the rest of the commits will be done as part of A4. Git Portfolio.

Next, you'll be asked to clone the provided data repository that will be the starting point for your project. You will also be asked to describe the initial form of the data you have cloned.

You will also be provided with an initial program in the file `main.py`. You will need to run this program and explain how your initial program understands and interacts with the information it receives from a user (input) and the results it provides back to the user (output). Finally, you will be asked to generate some ideas about future features you'd like to add to the program and how you might want to analyse this data and further data you'll locate and use yourself, as you move through the course.

The description of the data, the explanation of program behaviour with user inputs and outputs, and your ideas about future features and analysis will be done in a readme file called *readme.md*.

This can be found in the Assessments section on the course page. This will continue to be developed, along with your program over the course and submitted again with A.2 Project part 2 – Mini-project and A.3 Project part 3 – Final project.

The readme file will need to be submitted both through Git as well as through the course site. The course site is where your assignment will be marked, and the feedback will be provided. We're asking you to also submit the `readme.md` through Git because this is where readme files will be logged in the real world, and using this platform effectively is a vital real-world skill.

Note: All the elements that make up the project and assessments must only be pushed to the server that is provided for this course.

Late Submission Conditions

Submission of the assessment deliverables will be accepted up to one week (7 days) from the due date listed for this assessment.

Late submissions are subject to a 15% raw mark deduction.

Example:

Vesper completed an assignment worth 80 marks in total. Their submission once marked scored 75 marks out of 80. Vesper submitted their assignment three days late; thus, a further 12 marks were deducted – 15% of the available marks – Vesper receives 63 out of the 80 available marks.

Practical application

Selecting and cloning datasets is essential for data scientists. Additionally, the ability to use Git is crucial for those involved in programming, as it is a widely used tool for tracking software development. Finally, understanding user interactions with a program is vital for creating effective programs.

Instructions

1. Access your Git repository

- a) A Git repository has been created for you:
 - i. You can find out how to access this by completing the Git exercises in Module 1.

2. Inspect the project files

- a) Download and inspect the files for the project:
 - i. `main.py`

- ii. `readme.md`

- b) These files can be found in the Assessments section on the course page under A1. Project part 1 – Read-me.

3. Install Git.

- a) Follow the directions for installing Git that can be found on the course page under Module 0: Software installation.

4. Set up your repository

- a) Clone the COSC480 project repository.

- i. Use the same method covered in the Git Exercises found in Module 1 on the course page.

- b) Add initial project files.

- i. Place the files that compose your selected initial project concept into the folder you will use for this project.

- ii. Commit these files with an appropriate commit message.

- c) Fill in the missing sections of **readme.md**.

- i. This file can be found on the course page under A1. Project part 1 – Read-me in the Assessments.

- ii. Complete all sections except Dependencies and Student Lead features which will be completed in later assignments.

- d) Commit the updated `readme.md` file with an appropriate commit message.

- e) Push your changes to your repository.

Recommended structure

Please follow the structure set in readme.md:

1. Initial Program Behaviour:
 - Explanation of initial data form.
 - Program behaviour regarding user inputs and outputs.
2. Dependencies:
 - No need to complete for this assessment.
3. How to Run:
 - Already completed.
4. Future Development:
 - a) Basic Analysis Feature:
 - Explain a minimum of one analysis you plan to run.
 - b) Student lead features:
 - Outline features you want to include.
 - This should be beyond the minimum features outlined in A2. Project part 2 – Mini-project outlined below:
 - Allow users to select year for monthly rain report.
 - Warn the user if no records found.
 - Create bar chart representation of monthly rainfall.
 - Add *Select all stations* option.

Research and referencing

Any ideas that are not your own need to be cited [using APA 7th edition](#).

Presentation guidelines

Work is expected to be word-processed and submitted using the provided template (readme.md) and be within 10% of the given word count (excluding any referencing). Spellchecking and proofreading of work before submission is strongly encouraged.

Generative AI Tools Are Not Restricted for This Assessment



In this assessment, you are permitted to use generative artificial intelligence (AI) to assist you in any way within the bounds of academic integrity. All work submitted must be substantially your own and any work generated in part or in whole using AI must be cited, with the inclusion of prompts used and replies generated.

You must clearly indicate which AI tools were used and how they contribute to your assessment in an AI declaration attached as an appendix to your submitted work.

Assessment rubric

Level	0	1
Readme submitted to Git	•Readme not submitted.	•Readme successfully submitted.
Points	0 points	1 point

Level	0	1	2
Commit message	•No commit message submitted.	•Commit message was not sufficiently descriptive. •Commit message was overly lengthy and should have been broken down into smaller sub-units of work.	•Commit message descriptive. •Commit message was of appropriate length and does not require being chunked further.
Points	0 points	1 point	2 points
Initial Program Behaviour	•No description of the data. •No explanation of program behaviour present.	•Data description is vague or incomplete. •Explanation of program behaviour unclear or incomplete.	•Data description includes all essential features. •Explanation includes all essential features about inputs and outputs.
Points	0 points	2 points	4 points
Dependencies	•No list of libraries presented.	•Some libraries needed to run the program are listed but some are missing.	•All libraries need to run the program are listed.
Points	0 points	1 points	2 points
Future development	•No ideas for future analysis are provided. •No features for development are provided.	•Ideas for future analysis are vague or lack feasibility. •Features for development are vague or lack feasibility.	•Ideas for future analysis include all essential features and are feasible. •Features for development are clearly defined and feasible.
Points	0 points	2 points	4 points